Active Directory Lab (VMware)

Domain Deployment and Migration

Kale Dunlap
March 18, 2026

---

1. Project Overview

This project demonstrates the deployment, expansion, and migration of an Active Directory environment using VMware. The lab simulates a real-world enterprise scenario where a legacy domain controller is upgraded to a newer server version, followed by decommissioning of the original system.

---

2. Lab Architecture

Environment Components

- Domain Controller 1 (DC01)
    - Windows Server 2022
- Domain Controller 2 (DC02)
    - Windows Server 2025
- Client Machine (CLIENT01)
    - Windows 10/11

Starting Network Configurations

- VMware Network Adapter = NAT
- DC-01:
    - IPv4 Address: 192.168.88.10
    - Gateway: 192.168.88.2

    - Subnet: 255.255.255.0
    - DNS: 127.0.0.1
- DC-02:
    - IPv4 Address: 192.168.88.11
    - Gateway: 192.168.88.2
    - Subnet: 255.255.255.0
    - DNS: 192.168.88.10
- Client-01:
    - IPv4 Address: 192.168.88.130
    - Gateway: 192.168.88.2
    - Subnet: 255.255.255.0
    - DNS: 192.168.88.10

Domain Details

- Domain Name: corp.local
- Initial Functional Level: Windows Server 2016

---

3. Phase 1 – Initial Domain Deployment

Description

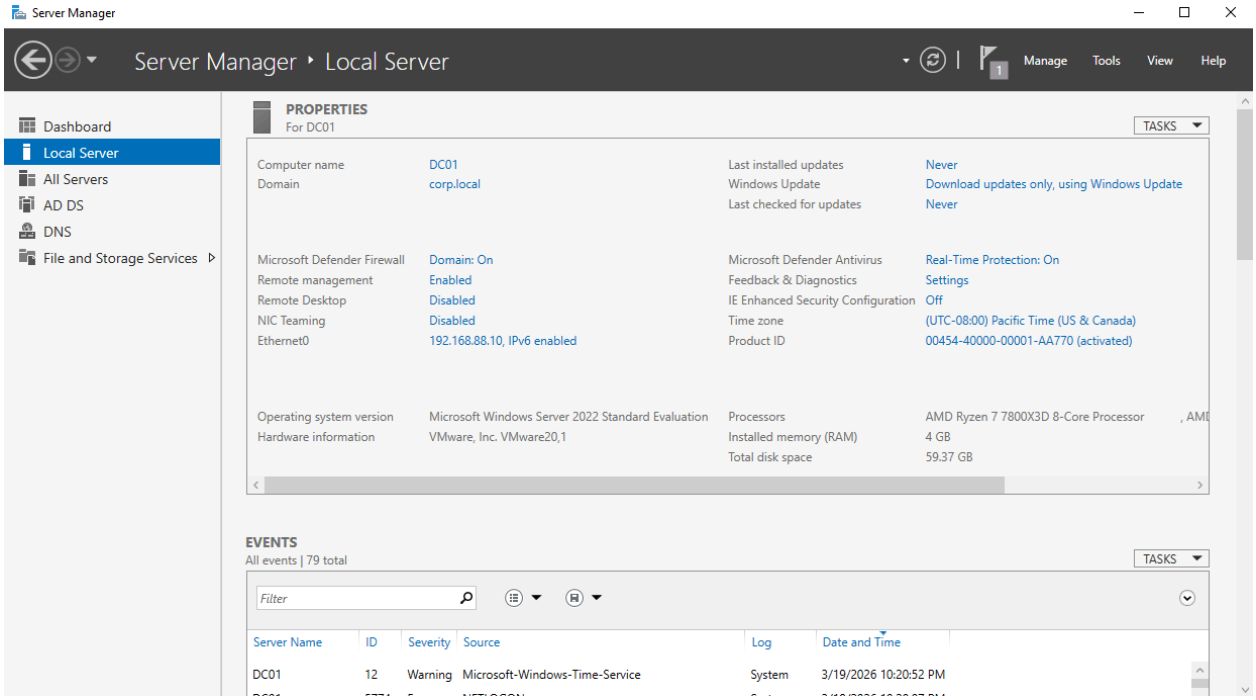
A new Active Directory forest is created using Windows Server 2022. This server acts as the initial domain controller for the environment.

Key Outcomes

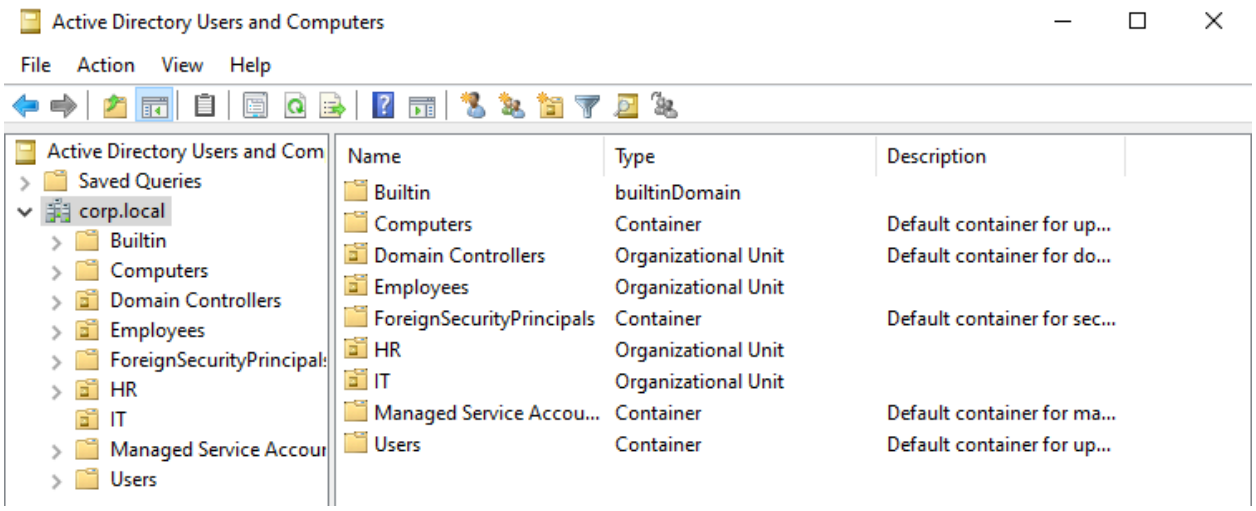
- Domain controller (DC01) deployed
- New forest (corp.local) created
- Functional level set to Windows Server 2016

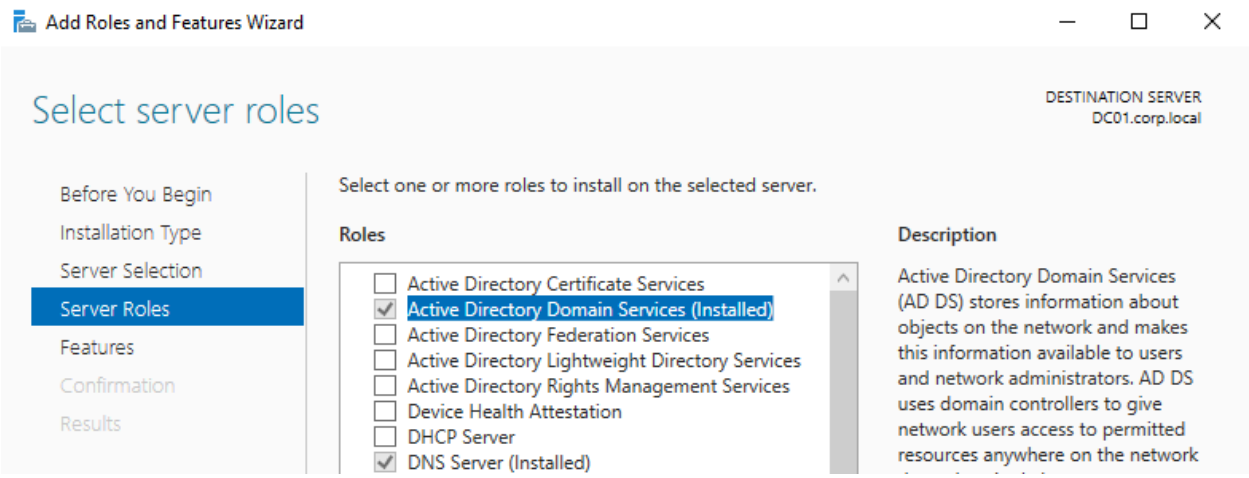
Screenshots

DC-01 Server Dashboard:

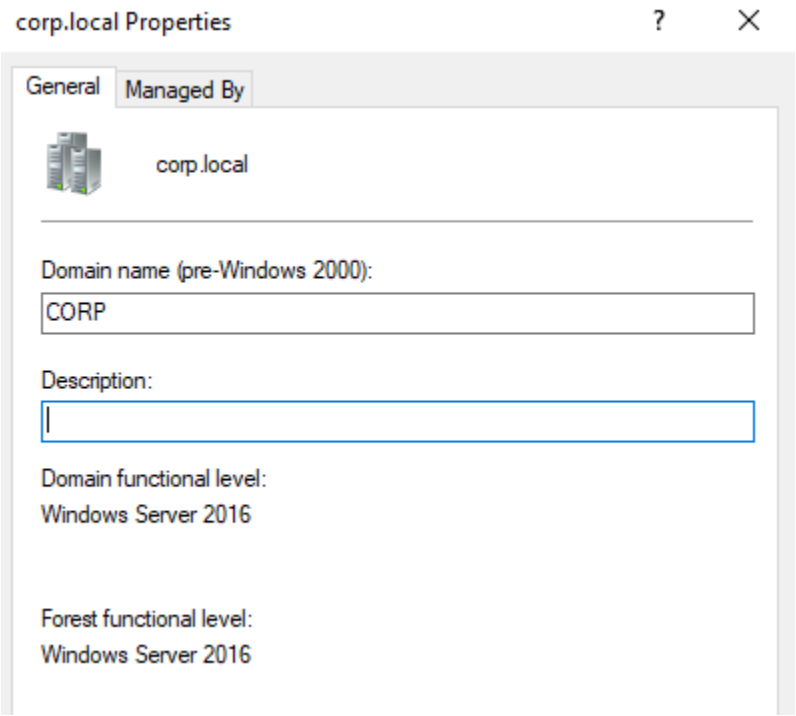


AD:





DC-01 functionality info:



---

4. Phase 2 – Active Directory Configuration

Description

The domain structure is configured to simulate a small enterprise environment with users, groups, and organizational units.

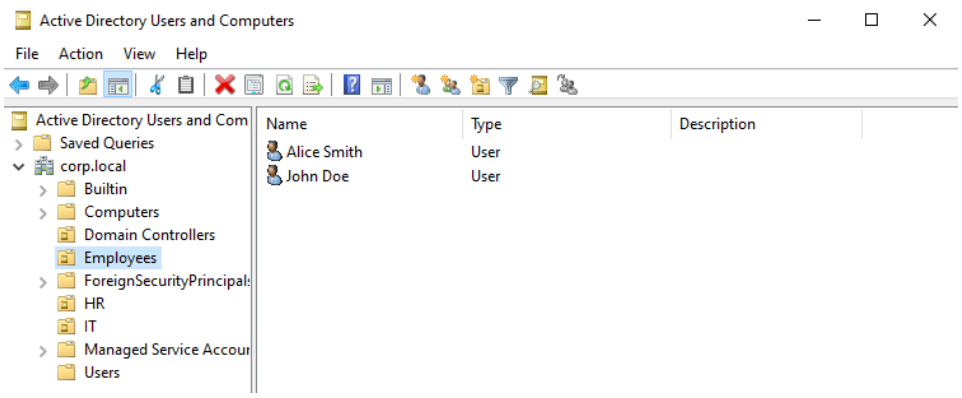
Key Outcomes

- Organizational Units created (IT, HR, Employees)
- Users provisioned (jdoe, asmith, itadmin, CKLozano)
- Security groups configured and assigned

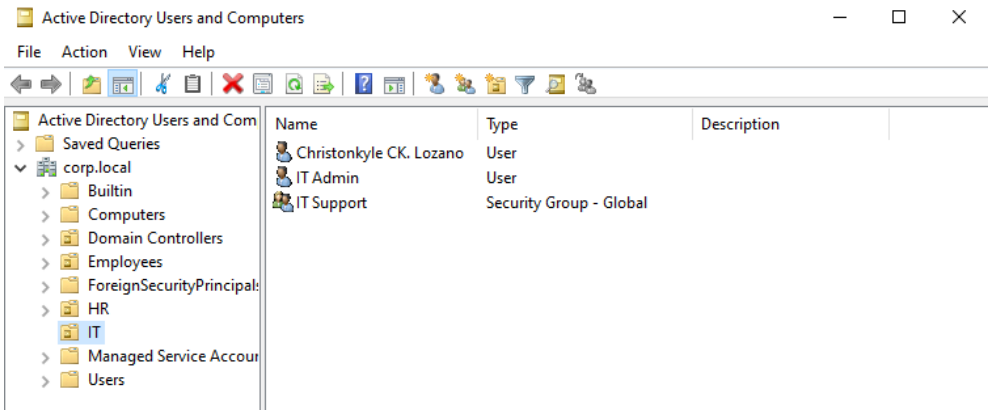
Screenshots

Active Directory Users and Computers (OU structure):

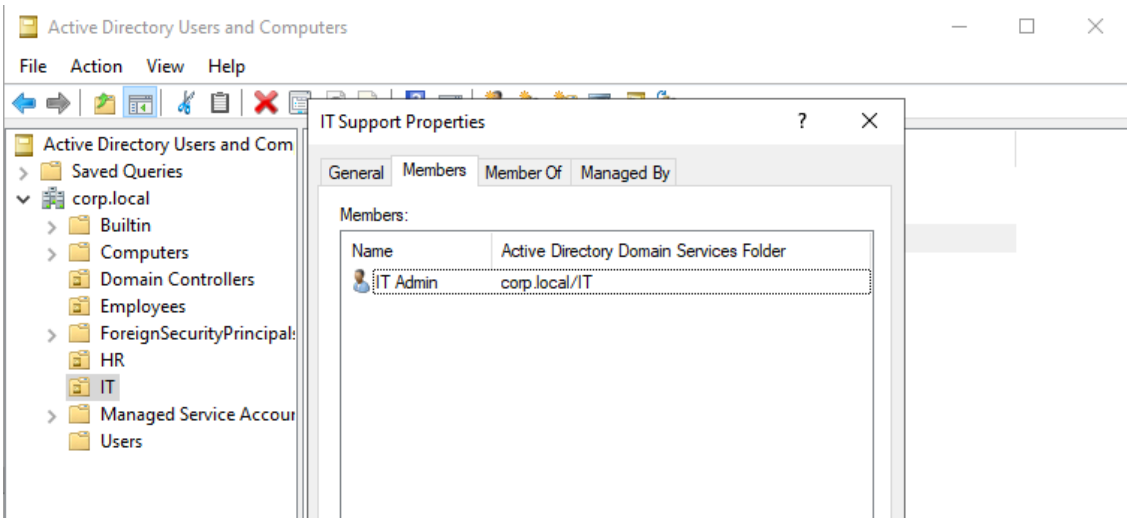
Employee OU:



IT OU + Security Group:



Group membership configuration:



---

5. Phase 3 – Client Integration

Description

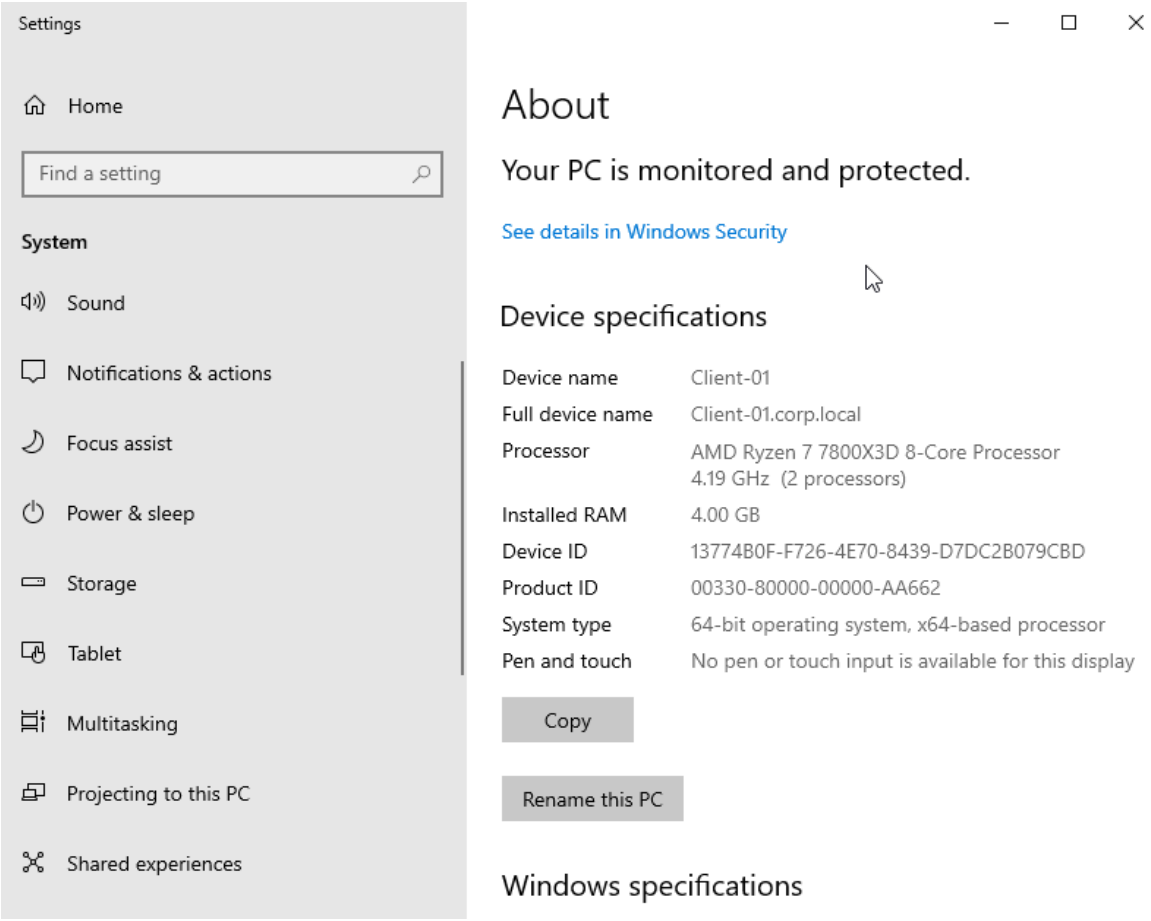
A client machine is deployed and joined to the domain to simulate an end-user workstation in the environment.

Key Outcomes

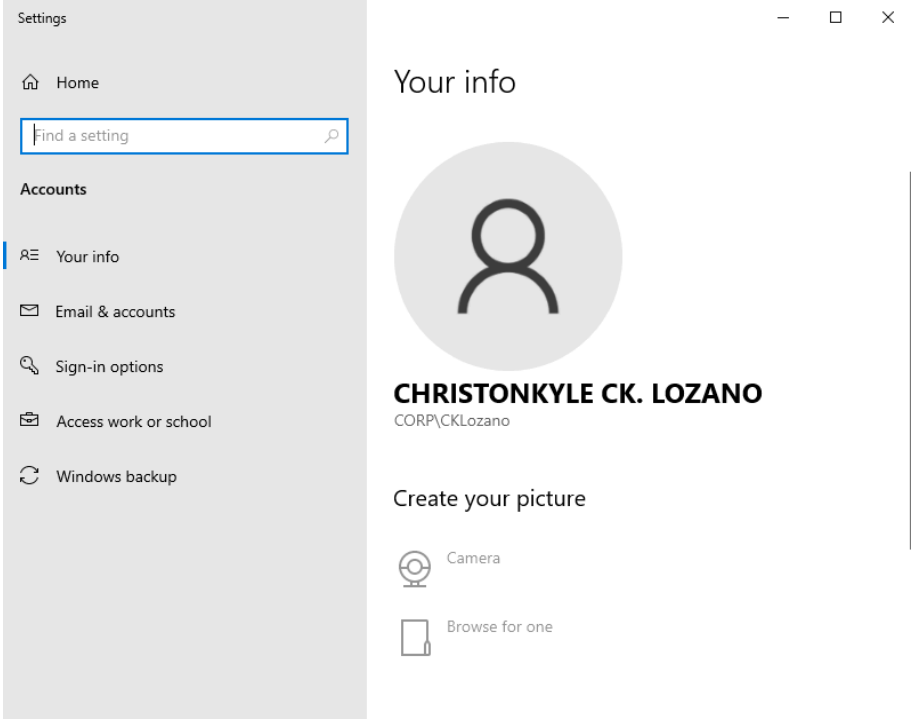
- CLIENT01 successfully joined to corp.local
- Domain user authentication verified

Screenshots

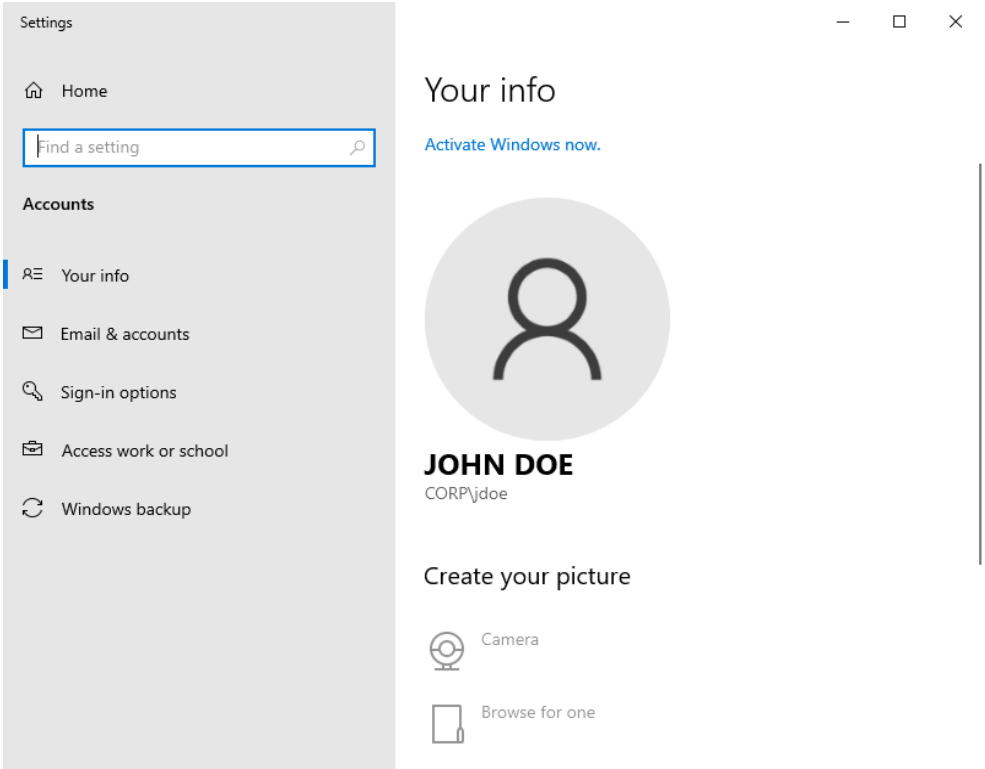
Client-01 joining domain (corp.local):



‘CKLozano’ user sign-on:



‘jdoe’ user sign-on:



6. Phase 4 – Domain Expansion (Windows Server 2025)

Description

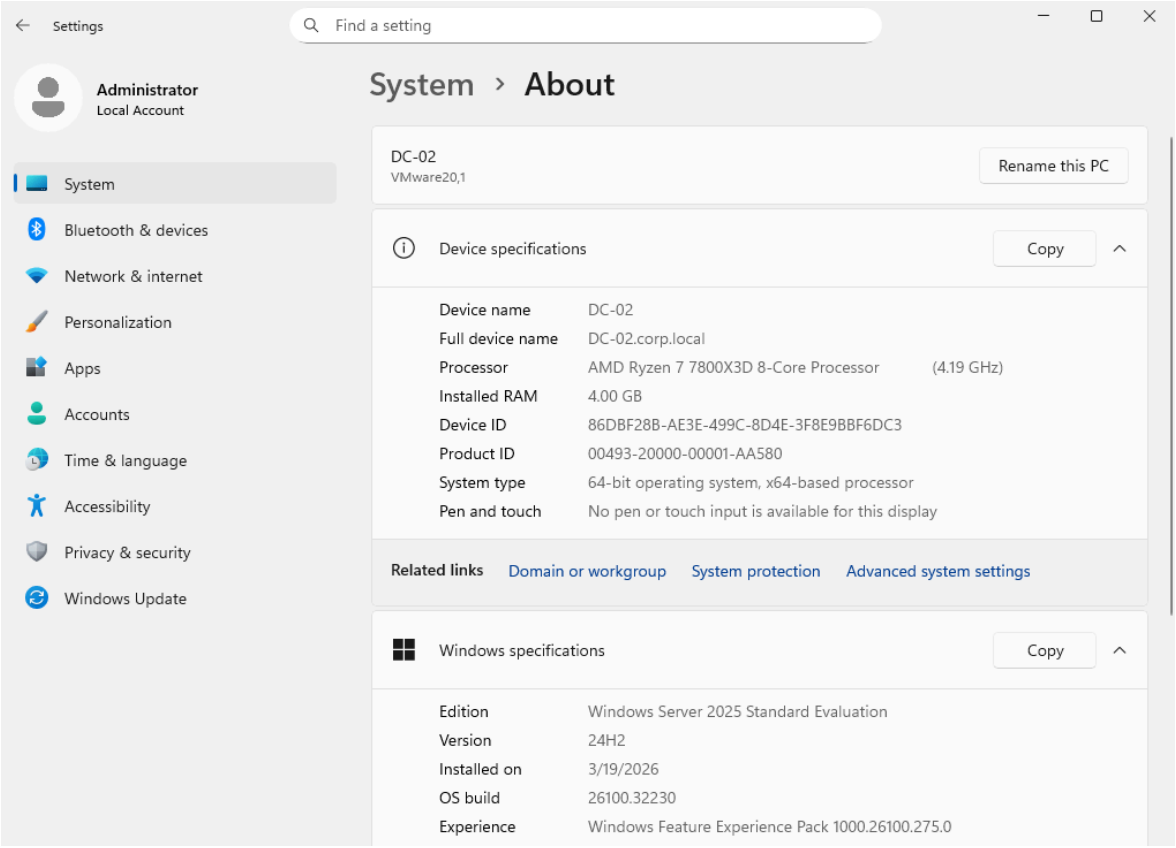
A second domain controller is introduced using Windows Server 2025 and joined to the existing domain to support migration and redundancy.

Key Outcomes

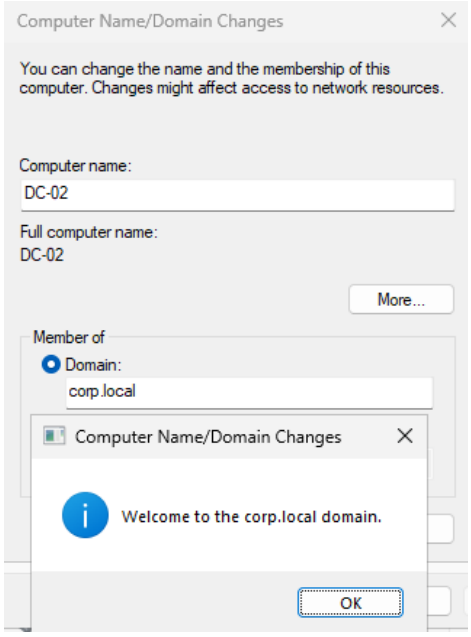
- DC-02 joined to corp.local
- Active Directory Domain Services installed
- DC-02 promoted to domain controller

Screenshots

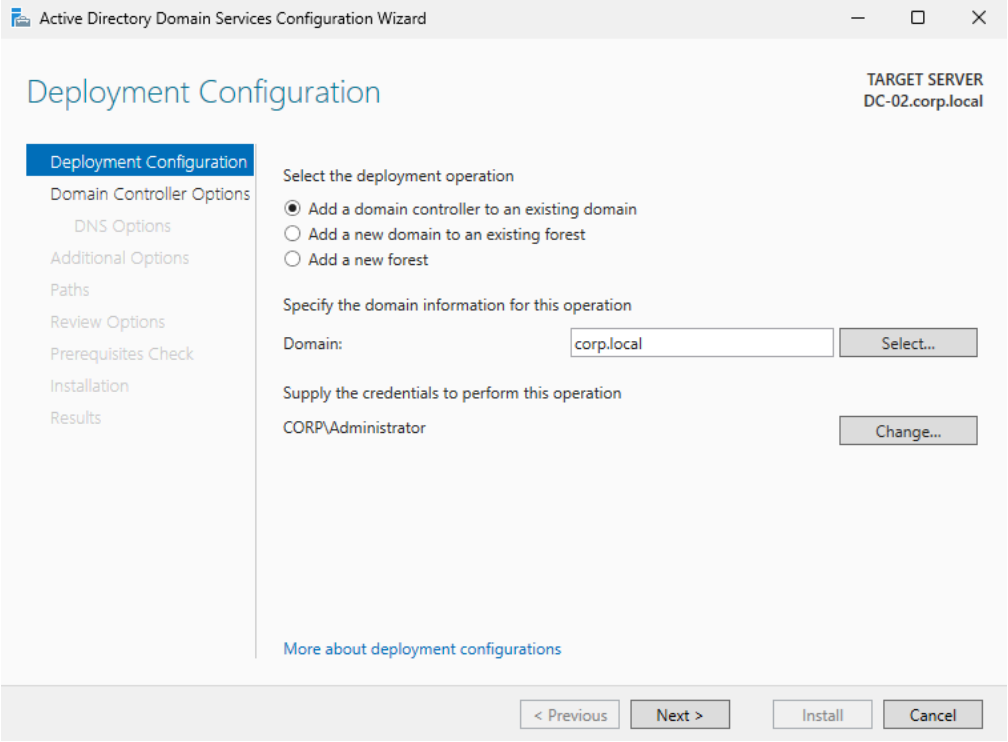
DC-02 system info:



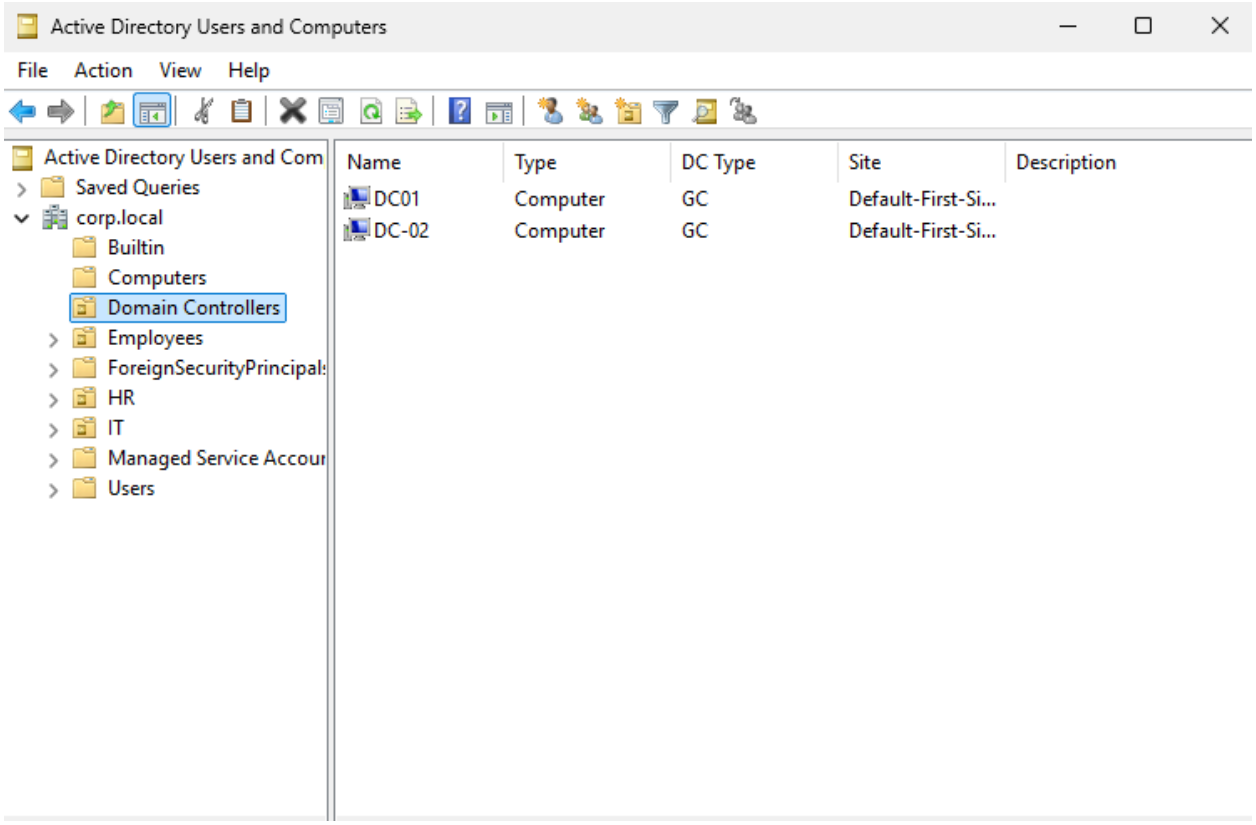
Domain (corp.local) join confirmation:



Promoting to DC (adding to corp.local):



AD showing both DCs:



---

7. Phase 5 – Active Directory Migration and Upgrade

Description

The environment is transitioned from the original domain controller to the new server. Core roles and services are transferred, and the domain is upgraded to a newer functional level.

Key Outcomes

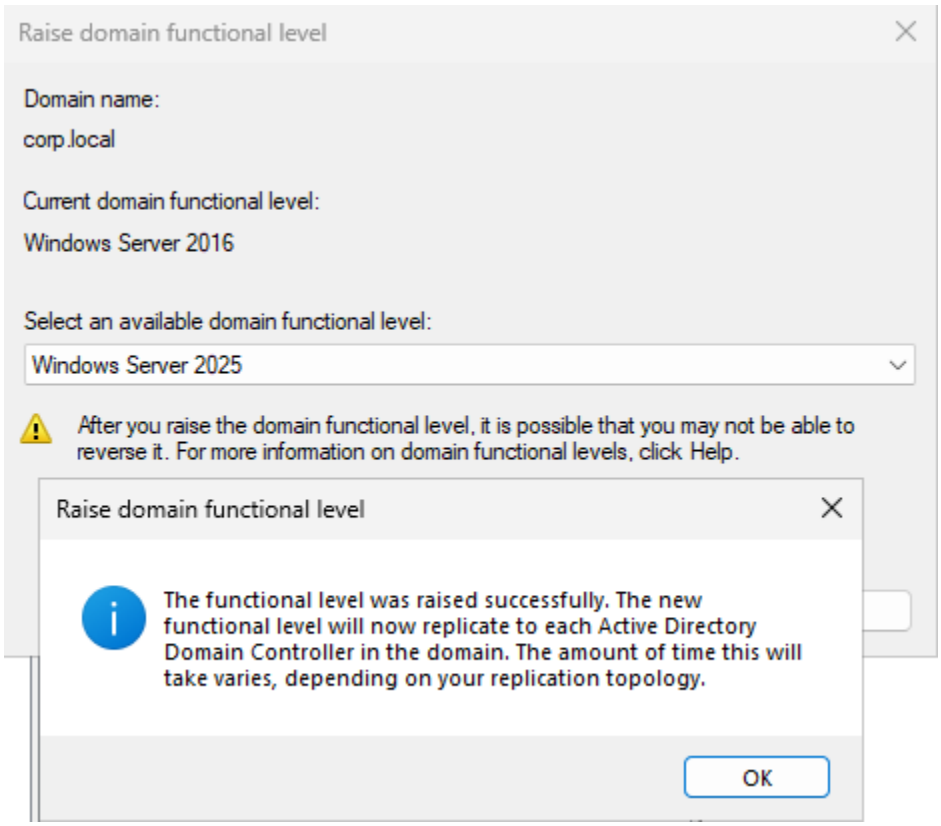
- FSMO roles transferred to DC02
- DNS services operating from DC02
- Domain and forest functional levels raised to Windows Server 2025

Screenshots

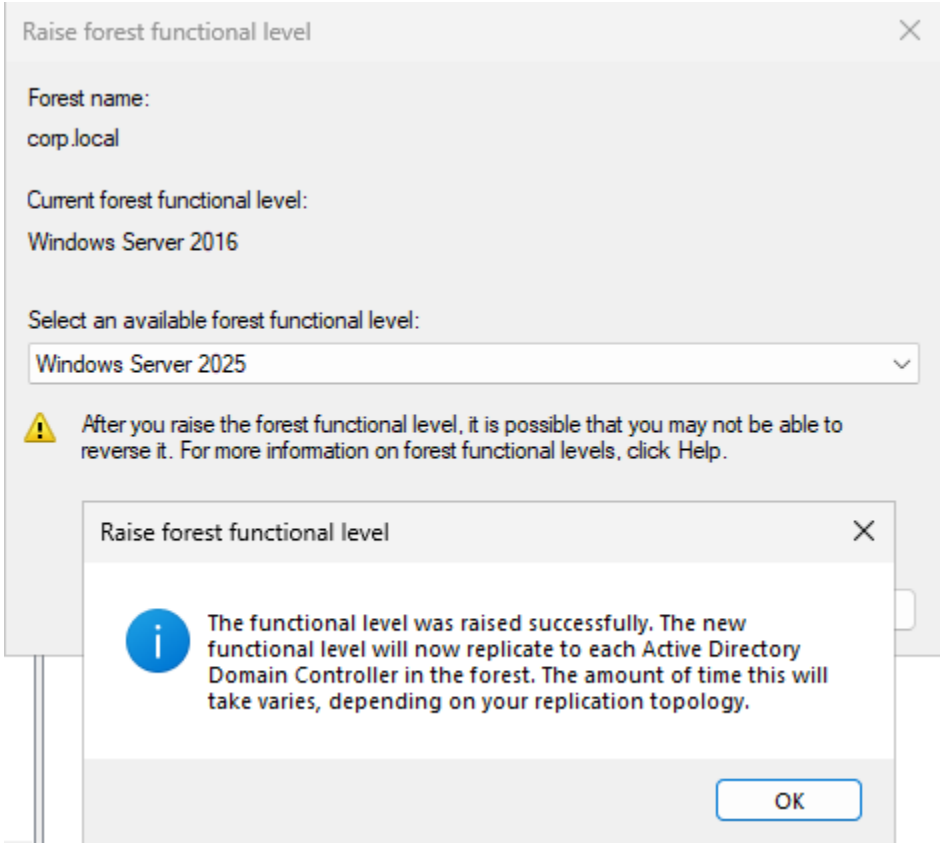
FSMO role transfer confirmation:

```
C:\Users\Administrator.CORP>netdom query fsmo
Schema master               DC-02.corp.local
Domain naming master        DC-02.corp.local
PDC                         DC-02.corp.local
RID pool manager            DC-02.corp.local
Infrastructure master       DC-02.corp.local
The command completed successfully.
```

Domain upgrade to Windows Server 2025:



Forest Upgrade to Windows Server 2025:



---

8. Phase 6 – Decommissioning Legacy Domain Controller

Description

The original domain controller (DC01) is safely removed from the environment after confirming full migration to DC02.

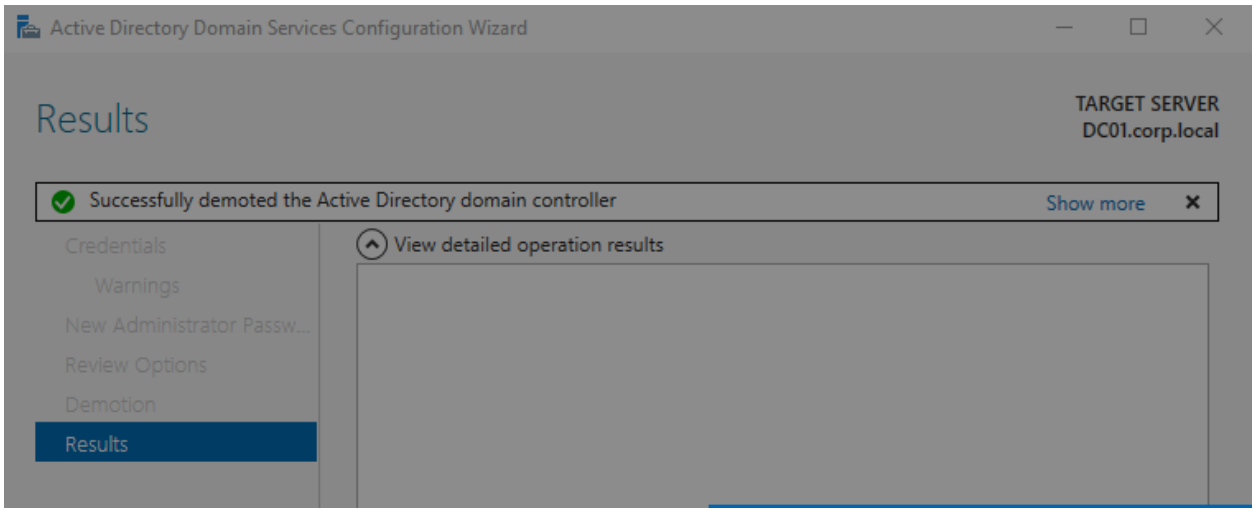
Key Outcomes

- DC01 demoted from domain controller

- AD DS role removed
- DC01 removed or powered off

Screenshots

Demotion wizard



- AD showing only DC02
- Final server list

---

9. Phase 7 – Validation and Testing

Description

The environment is tested to ensure stability and functionality after migration and decommissioning.

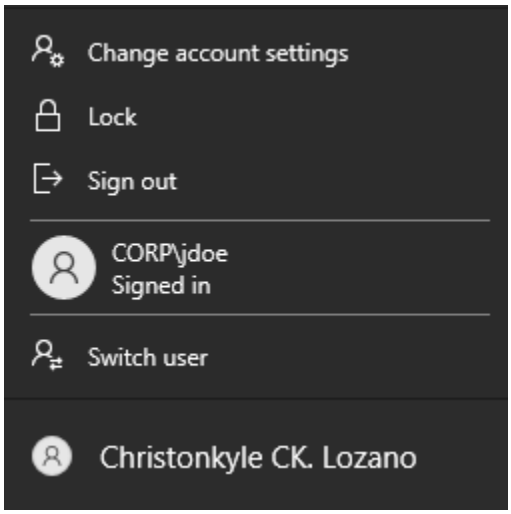
Key Outcomes

- Domain authentication successful
- Users and groups intact

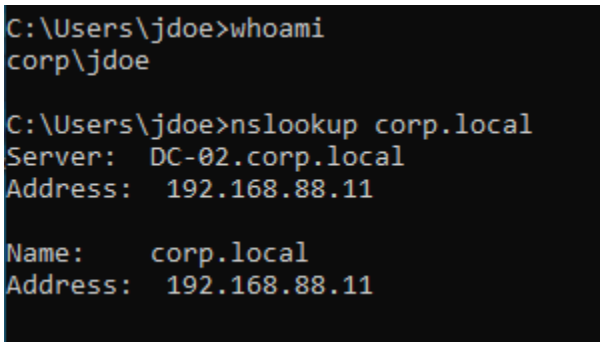
- DNS resolution functioning
- Client connectivity verified

Screenshots

Successful domain login (logged in as both CK / jdoe) on Client-01:



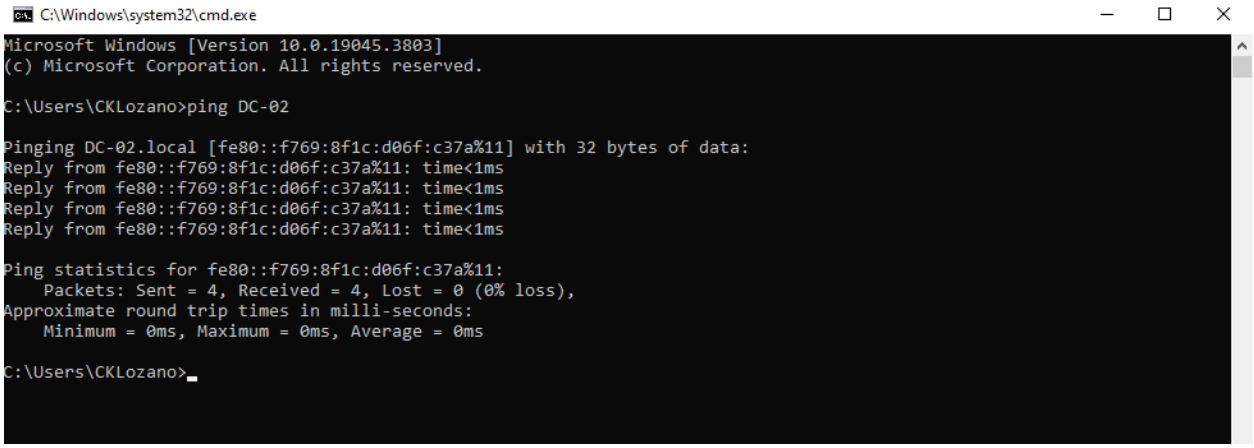
Network/DNS validation:



```
C:\Users\jdoe>whoami
corp\jdoe

C:\Users\jdoe>nslookup corp.local
Server:  DC-02.corp.local
Address:  192.168.88.11

Name:    corp.local
Address:  192.168.88.11
```



Overview:

This lab demonstrated the deployment and migration of an Active Directory environment in VMware. A domain was created, users and organizational units were configured, and a client machine was joined to validate authentication. A second domain controller was introduced, FSMO roles were transferred, and the original controller was decommissioned. The environment was upgraded and tested, confirming proper DNS resolution, connectivity, and overall domain functionality.